

HW2

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```
suppressPackageStartupMessages(library(dplyr))
dt <- read.table('cult_data.dat', header = FALSE)
colnames(dt) <- c('field', 'cultivar', 'soil', 'weight')
dt <- dt %>%
  group_by(soil, cultivar) %>%
  mutate(group = cur_group_id()) %>%
  ungroup()
dt <- dt %>%
  mutate(across(c(field, cultivar, soil, group), as.factor))
```

1

```
anova(lm(weight~cultivar*soil, data = dt))
```

Analysis of Variance Table

Response: weight

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
cultivar	1	2.407	2.407	1.0013	0.3303
soil	2	118.176	59.088	24.5830	7.13e-06 ***
cultivar:soil	2	1.826	0.913	0.3798	0.6893
Residuals	18	43.265	2.404		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

According to the ANOVA output, when testing just the cultivar and soil effects (and their interaction), cultivar and the interaction do not have a significant effect on the growth of the plants. However, we do see that the soil has a significant effect.

2

```
y <- dt$weight
w <- model.matrix(y~group-1,data=dt)
muhat <- solve(t(w)%*%w)%*%t(w)%*%y
muhat
```

```
      [,1]
group1 27.900
group2 28.025
group3 29.250
group4 30.650
group5 33.150
group6 33.525
```

```
# Group 1: Control, A
# Group 2: Control, B
# Group 3: Dead, A
# Group 4: Dead, B
# Group 5: Live, A
# Group 6: Live, B
```

3

```
y <- dt$weight
w <- model.matrix(y ~ group-1, data=dt)
x <- model.matrix(y ~ group, data=dt)

#           | Cultivar a | Cultivar b |
# Soil Control |    1      |    2      |
# Soil Dead   |    3      |    4      |
# Soil Live   |    5      |    6      |

c <- rbind(
  c(1/6,1/6,1/6,1/6,1/6,1/6), # Overall mean
  c(1/2,1/2,-1/2,-1/2,0,0), # Soil Control vs. Soil Dead
  c(0,0,1/2,1/2,-1/2,-1/2), # Soil Dead vs. Soil Live
```

```

c(1/3,-1/3,1/3,-1/3,1/3,-1/3), # Cultivar A vs. B
c(1,-1,-1,1,0,0), # Interaction 1
c(0,0,1,-1,-1,1) # Interaction 2
)

```

```

muhat <- solve(t(w)%*%w)%*%t(w)%*%y
beta <- solve(t(x)%*%x)%*%t(x)%*%y
sse <- t(y-x)%*%beta)%*%(y-x)%*%beta)
mse <- as.numeric(sse/(24-6))
varmuhat <- mse*solve(t(w)%*%w)

```

```

c_cult <- c[4, ]
fhat <- t(c_cult)%*%muhat)%*%solve(c_cult)%*%varmuhat)%*%c_cult)%*%muhat/1
fhat

```

```

      [,1]
[1,] 1.001271

```

```
1-pf(fhat,1,23)
```

```

      [,1]
[1,] 0.327415

```

No, this test is not significant

4

```

c_soil <- c[2:3, ]
fhat <- t(c_soil)%*%muhat)%*%solve(c_soil)%*%varmuhat)%*%t(c_soil))%*%c_soil)%*%muhat/2
fhat

```

```

      [,1]
[1,] 24.58298

```

```
1-pf(fhat,2,22)
```

```

      [,1]
[1,] 2.464078e-06

```

Yes, this test is significant

5

```
c_int <- c[5:6, ]  
fhat <- t(c_int%%muhat)%%solve(c_int%%varmuhat%%t(c_int))%%c_int%%muhat/2  
fhat
```

```
      [,1]  
[1,] 0.3798105
```

```
1-pf(fhat,2,22)
```

```
      [,1]  
[1,] 0.6883895
```

No, there are not significant interactions

6

```
dt_red <- dt[!(dt$cultivar == 'b' & dt$soil == 'liv'),]  
  
w <- model.matrix(weight ~ group-1, data=dt_red)  
w1 <- w[,1:5] # Remove the all 0s column  
ym <- dt_red$weight  
muhat <- solve(t(w1)%%w1)%%t(w1)%%ym  
muhat
```

```
      [,1]  
group1 27.900  
group2 28.025  
group3 29.250  
group4 30.650  
group5 33.150
```

7

```

ai <- solve(c)
Xm <- w%*%ai
# Remove the second interaction (the one that depends on group 6) row to make it full rank
Xn <- Xm[,-6]

betahat <- solve(t(Xn)%*%Xn)%*%t(Xn)%*%ym
sse <- t(ym-Xn%*%betahat)%*%(ym-Xn%*%betahat)
mse <- as.numeric(sse)/(20-5)
varbeta <- mse*solve(t(Xn)%*%Xn)
newmuhat <- ai[,-6]%*%betahat
newmuhat

```

```

      [,1]
[1,] 27.900
[2,] 28.025
[3,] 29.250
[4,] 30.650
[5,] 33.150
[6,] 34.550

```

The estimate for the cell is 34.55, while the original with data was 33.525, so the estimate was pretty close (only ~ 1 away).

8

```

vestmu <- ai[,-6]%*%varbeta%*%t(ai[,-6])
cbind(newmuhat,sqrt(diag(vestmu)))

```

```

      [,1]      [,2]
[1,] 27.900 0.6506087
[2,] 28.025 0.6506087
[3,] 29.250 0.6506087
[4,] 30.650 0.6506087
[5,] 33.150 0.6506087
[6,] 34.550 1.1268873

```

Standard error of the estimated cell mean is ~ 1.13 , while the rest of the errors are about half that much, which makes sense. The error of the cell we estimated should be larger than the other estimates.